

Structural Engineering Laboratory

Parker Truss Design, Load Testing, and Failure Analysis

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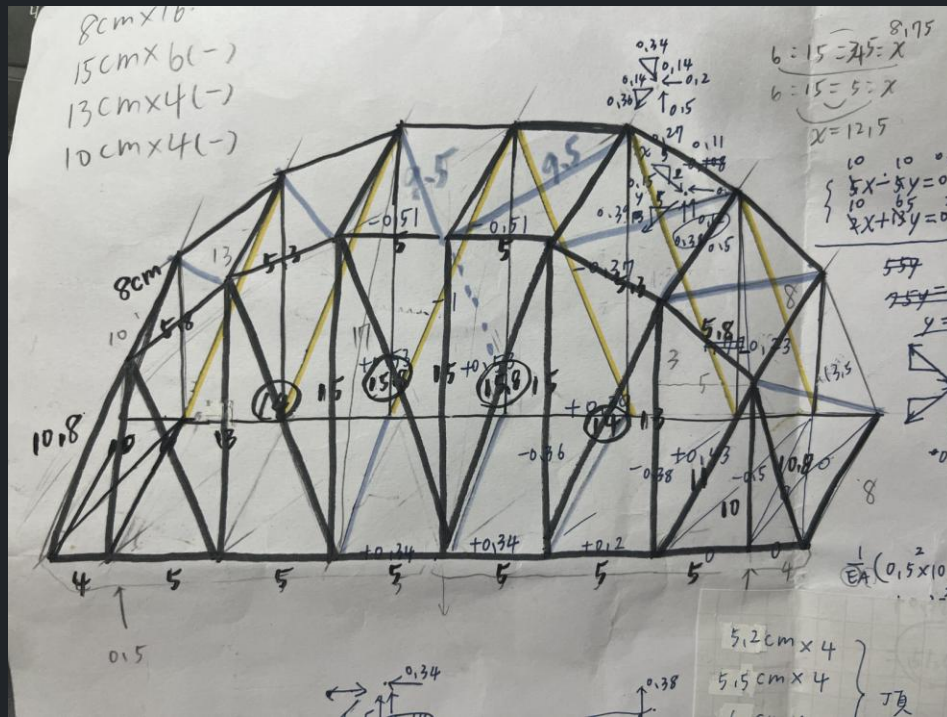
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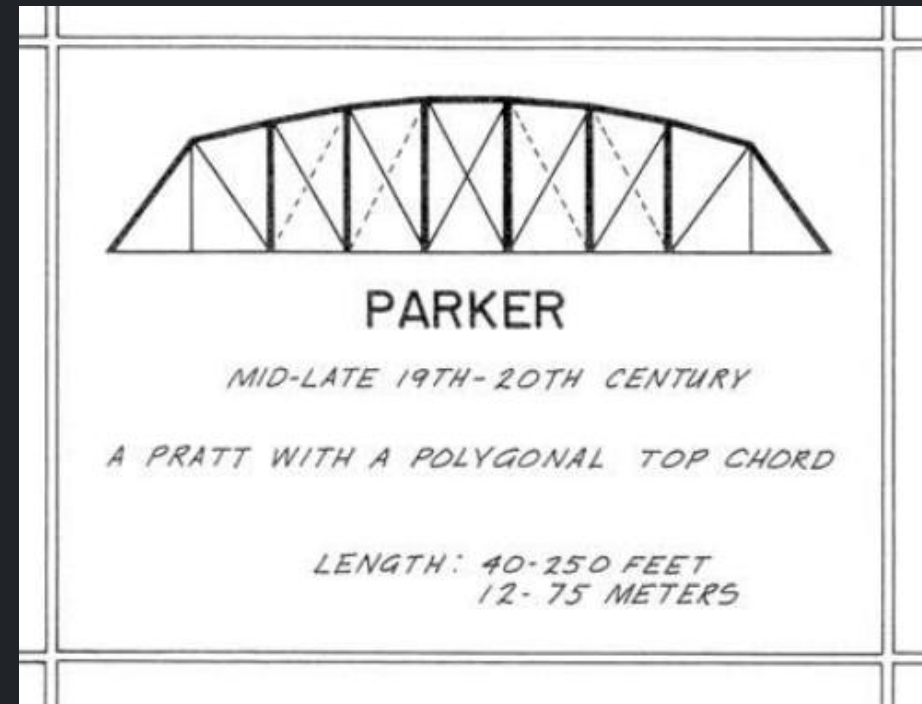
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Teaching Assistants: Yen-Kai Chen · Xing-Wen Tsai

Design Concept

The design was inspired by a Parker truss. To simplify the structural calculations, the bridge was configured as a statically determinate structure. The Parker configuration was selected to reduce material use while targeting a higher theoretical capacity. Based on preliminary estimate, this configuration could carry a higher load than a Pratt truss.



Design sketch



Design inspiration: Parker truss

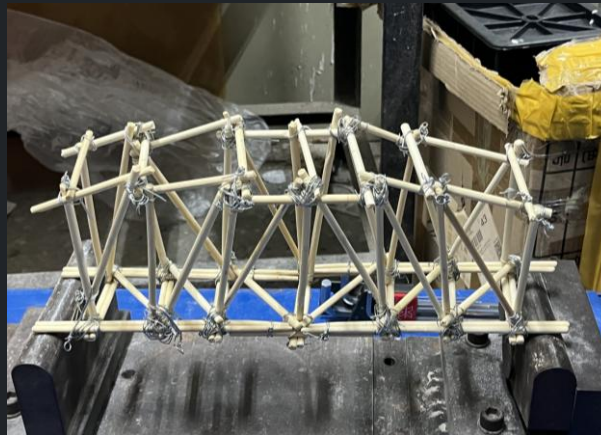
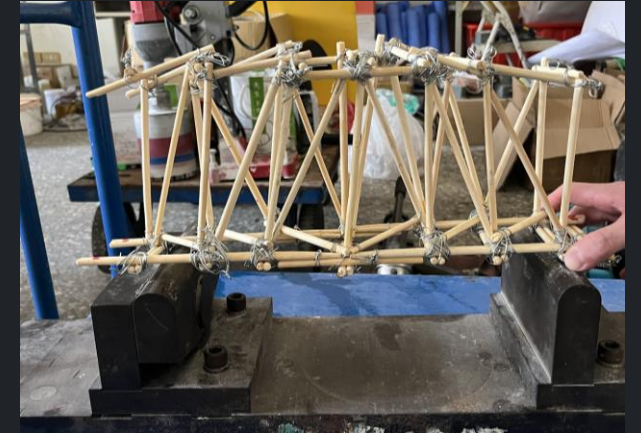
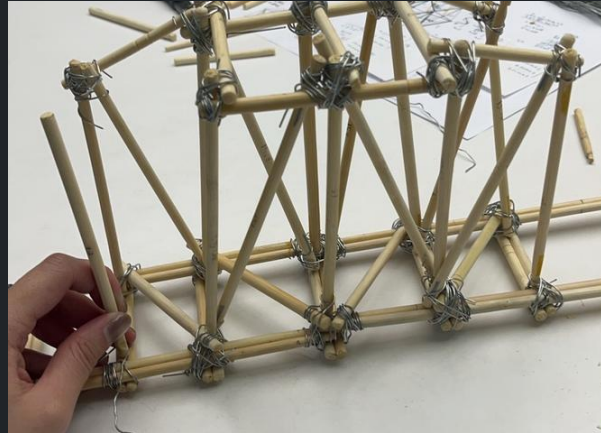
Fabrication Process

Construction sequence

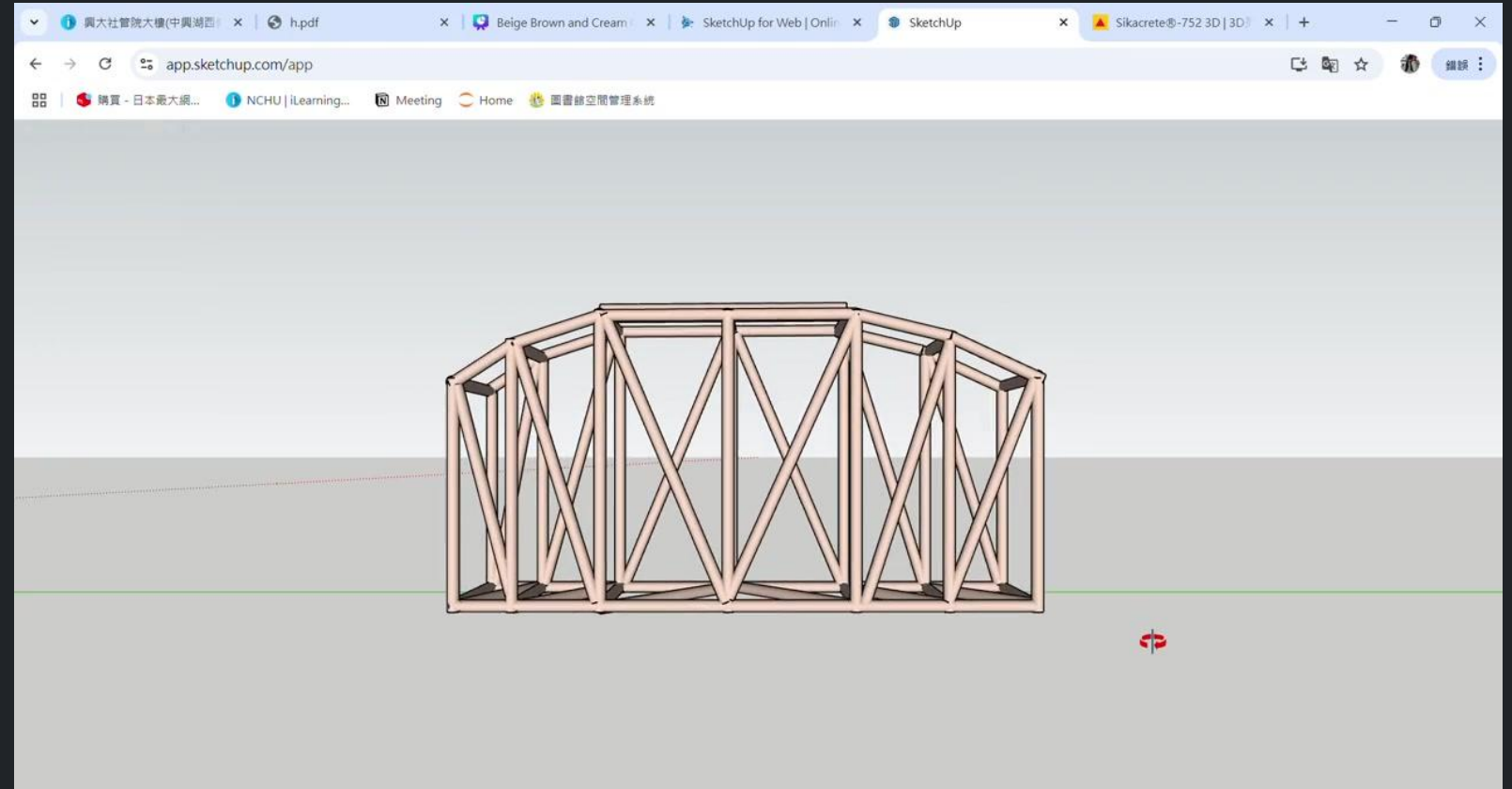
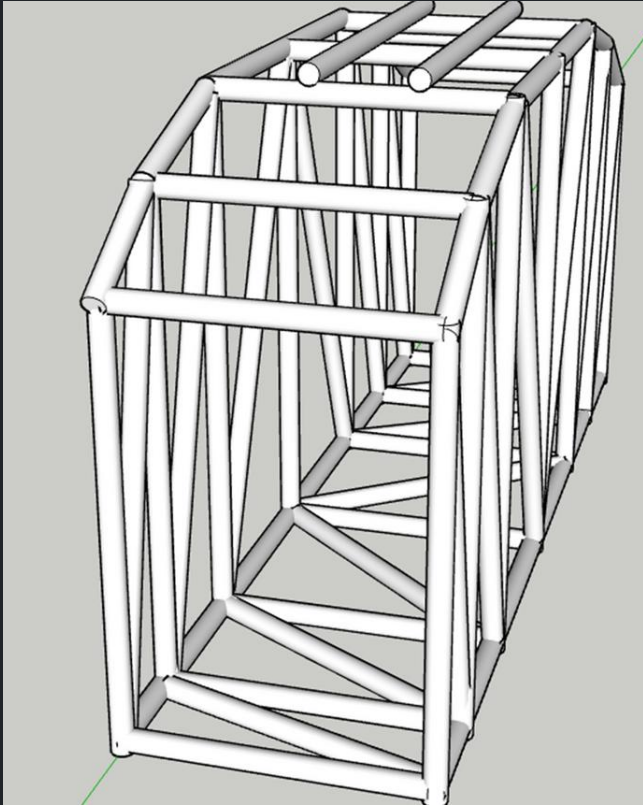
Bottom chord → build compression members outward from the center → install tension members

Problems and corrections

- Dimensional errors → insufficient extra length was reserved for wire tying.
- Overall structural skew → dimensions were not checked frequently enough, and members intended to be vertical were not kept truly vertical.
- Some joints loosened → the wire-tying method was modified.
- Weight reduction → zero-force members at both ends were removed, and top diagonal bracing was omitted.



3D Modeling of the Intended Geometry



A 3D model was created to visualize the intended geometry and compensate for imperfections in the fabricated prototype.

Preliminary Load Estimation I

Structural determinacy and virtual-work formulation

1. Check structural stability / determinacy

Members $m = 25$; joints $j = 14$; three support-condition equations. Using the determinacy relation shown in the original calculation:

m = members, r = support reaction component,
 j = joints

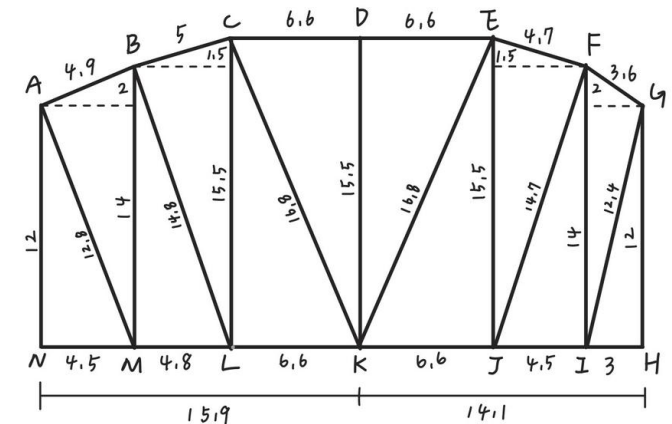
$$R = m + r - 2j \\ = 25 + 3 - 28 = 0$$

2. Use the unit-load method to relate displacement to external load

A virtual unit load is used to evaluate the displacement. The load is then adjusted until the displacement at the selected joint reaches 2 cm.

$$1 \times \Delta = \sum (n \cdot N \cdot L / EA)$$

Δ : displacement · n : member force from the unit virtual load · N : member force from the actual external load · E : bamboo elastic modulus · A : chopstick cross-sectional area



Truss dimensions used in the calculation

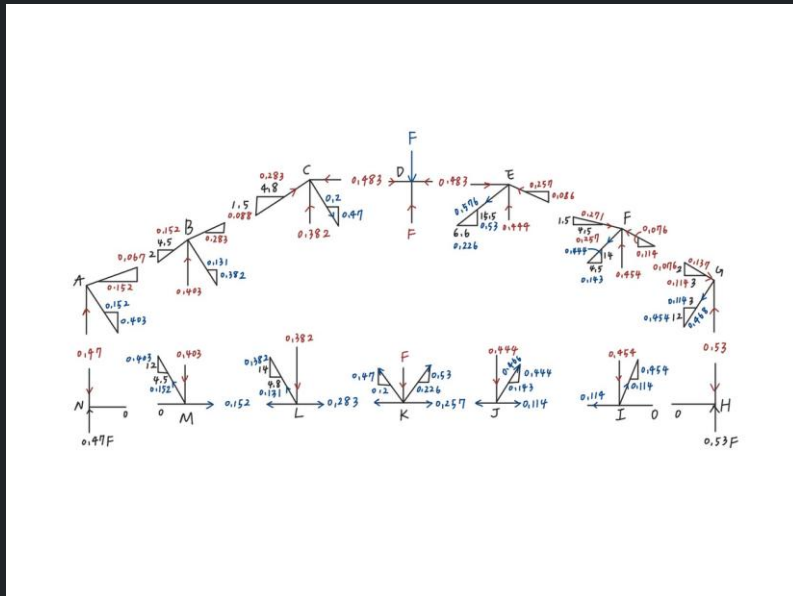
Preliminary Load Estimation II

Support reactions and member-force analysis

3. Assume an external load F is applied at point D and calculate the support reactions.

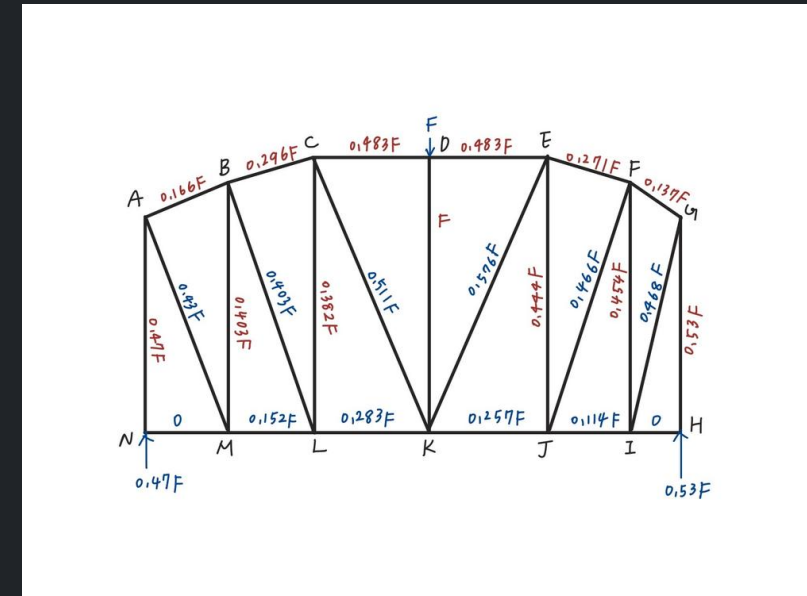
$$R_H = F \times 15.9 / 30 = 0.53F \quad R_N = F \times 14.1 / 30 = 0.47F$$

4. Calculate member forces at each joint.



Joint-by-joint static analysis under external load F

5. Assemble the member-force distribution under the actual external load.



Resulting member-force distribution

Calculation Notes

Sources of uncertainty in the analytical estimate

- Differences between the fabricated dimensions and the theoretical geometry.
- The wire-tied joints could loosen under load and deviate from the idealized joint behavior assumed in the analytical model.
- The elastic modulus was taken from reference data for bamboo and may not match the actual chopsticks.
- The 0.24 kg self-weight of the bridge was neglected in the original calculation.
- Members intended to be vertical were not perfectly vertical, and the overall structure was not perfectly symmetric.
- The bridge could fail before the midspan displacement reached the assumed 2 cm limit.

Experimental Results and Theory–Experiment Comparison

Theoretical capacity

20.10 kgf

≈ 197.1 N

Measured failure load

14.20 kgf

≈ 139.3 N

Measured / predicted

70.6%

29.4% below prediction

Difference: 5.90 kgf (≈ 57.9 N)

Interpretation

- The physical bridge failed at a lower load than the theoretical prediction.
- The observed gap is consistent with fabrication imperfections, joint looseness, uncertain material stiffness, geometric asymmetry, and modeling assumptions identified during the project.
- The test demonstrated the practical difference between idealized truss analysis and the behavior of a hand-built structure.